ABSTRACT

Eye diseases have been a significant health concern throughout history, with conditions such as Age-related Macular Degeneration (AMD) increasingly recognized as main causes of the vision impairment worldwide. Traditionally, the assessment of retinal conditions relied heavily on manual evaluations of Optical Coherence Tomography (OCT) images, which, while effective, are often time-consuming and require specialized expertise.

The Categorizing of retinal diseases utilizing OCT images has garnered considerable interest in the domain of computational medical imaging. This project introduces a deep learning methodology for the automated classification of OCT images via a custom Convolution Neural Network (CNN) in conjunction with recognized transfer learning models. Deep learning methodologies are favored for their exceptional accuracy and performance; however, they frequently necessitate substantial computational resources and time for training. We employ MobileNet, ResNet50, GoogleNet, and DenseNet for feature extraction and comparison analysis.

The models are trained and refined on an OCT dataset utilizing various hyperparameter settings, such as learning rate, epoch count and optimizing techniques to get maximal accuracy. Upon concluding our study, we evaluate the performance of these models and ascertain the optimal design based on accuracy and additional assessment criteria. Our methodology illustrates the capability of deep learning in the automated classification of OCT images, providing enhanced diagnostic support for the identification of retinal diseases.

One of the strongest components of this project is the comparison of multiple model classification performance metrics evalution, including accuracy, precision, recall, F1-score, and confusion matrices. By comparing different architectures in a structured way, we determine the best performing and accurate deep learning model for retinal detection of OCT images. The outcome of this research contributes to the literature in AI-assisted ophthalmology, offering a cost-efficient and scalable solution for early detection and screening of retinal disease. This project supports the Sustainable Development Goal (SDG) 3: Good Health and Well-being goal by promoting the utilization of new and advanced technology in healthcare in order to facilitate timely diagnosis, support patient outcomes, and improve the quality of healthcare services and practice. Additionally, with the automation of deep learning AMD detection, this project can support reducing preventable blindness through early detection of AMD and contribute to sustainable health practices.